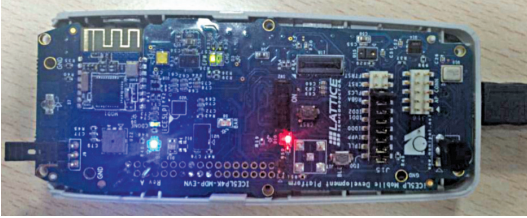
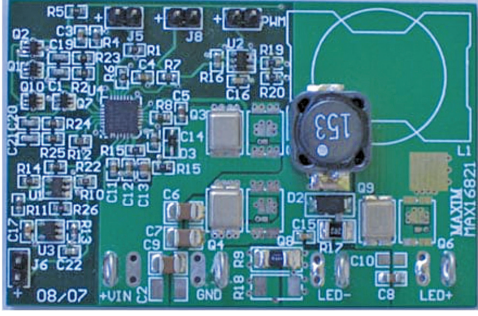


Interesting Reference Designs Of LED CONTROL SOLUTION

Title of Reference Design	RGB LED	6A RGB LED Driver
Image		
Key Attractions	The reference design offers comprehensive control over colour, brightness, and blinking via a user-friendly interface, which is suitable for rapid customisation in diverse applications.	The reference design for a 6A step-down LED driver in portable projectors efficiently manages RGB LEDs, ensuring constant current and rapid adjustments with minimal ripple for optimal lighting.
Highlights	The Lattice RGB reference design offers a low-cost option that incorporates an LED for any design, enabling adjustments to LED functions to meet specific application requirements. This design provides control over an RGB LED's colour, brightness, intensity, blink rate, and breathing dynamics through a software GUI, compatible with MAC and Windows PCs via USB. It allows control of LED functionalities and testing on a few hardware evaluation boards. The iCE40 Ultra mobile development platform operates multi-colour LEDs with individual brightness settings through pulse width modulation (PWM), automatic blinking, and an optional breath on/off feature, powered by an iCE40 Ultra device. This device features an I2C slave interface, data control, buffer, and LED control functionalities, and is paired with an RGB driver that includes a constant current sink and a power-on reset function.	Analog Devices' reference design for a 6A step-down LED driver, intended for portable projectors, uses the MAX16821, a PWM LED driver, to power a single LED with three units needed for RGB LEDs. The design operates from a 10V to 15V supply, stepping down to a constant current for an LED with a forward voltage range of 4.5V to 6V. Implemented with a switching frequency of 300kHz set by a 200kΩ resistor, the system manages a steady current through regulation of inductor current, matching the LED current. This configuration uses dual control loops: an inner current loop that adjusts based on the outer voltage loop's output, and an outer voltage loop that sets the inner loop's regulation parameters. Furthermore, the design allows the LED current to adjust within 1μs to changes when set at 6A during PWM, using a small output capacitor and a larger inductor to maintain low current ripple and stable inductor current at 6A.
Applications	The reference design can be interfaced with the application processor to drive the LED effectively.	The reference design is suitable for applications such as projectors, large-scale displays, and advanced lighting setups.
OEM Brand	Lattice Semiconductor	Analog Devices (ADI)
URL	https://www.electronicsforu.com/electronics-projects/rgb-led-reference-design	https://www.electronicsforu.com/electronics-projects/6a-rgb-led-driver-reference-design